

Proof. We construct $\mathcal{B}$ via a greedy algorithm similarly to the Gilbert-Varshamov bound. Start with $\mathcal{B} = \emptyset$ and add any $B \subset [nK]$ of size K not already in $\mathcal{B}$ to $\mathcal{B}$. Each new element blocks at most

$$\binom{K}{cK} \binom{nK - cK}{K - cK}$$

elements from being added to $\mathcal{B}$. Hence the maximum size of $\mathcal{B}$ is at least

$$\binom{nK}{K} \binom{K}{cK}^{-1} \binom{nK - cK}{K - cK}^{-1} = \binom{nK}{K} \binom{K}{cK}^{-2} \geq O(n^K K^{-1/2}) \cdot 2^{-2K} \geq e^{\Omega(K)}$$

for $n \geq 5$. □

Then $|E_s(p_B) \cap E_s(p_{B'})| = |B \cap B'| \leq cK$ and we can repeat the same argument to show orthogonality. Furthermore by taking r sufficiently large, the local neighborhood of any path contained in the bottom horizontal line (which must be contained in the dashed area in Figure 3) is isomorphic for all $B \in \mathcal{B}$, since it cannot query the endpoint states to identify which clusters are actually connected. Hence $\{p_B : B \in \mathcal{B}\}$ satisfies the conditions to compute the SQ dimension with access to $\text{nbd}(\mathcal{M})$, and thus

$$\text{SDA}(\mathcal{P}; \text{nbd}(\mathcal{M})) \geq |\mathcal{B}| \geq e^{\Omega(K)}.$$

□